



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2020 Year 11 Course Assessment Task 3

Tuesday, 30 June 2020

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Commence each new question on a new Writing Booklet. Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, arrange the booklets used in the correct order after this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 11MAX.1 – Ms Madden
- ☐ 11MAX.2 – Mr Siu
- ☐ 11MAX.3 – Ms Ham
- ☐ 11MAX.4 – Mr Sekaran
- ☐ 11MAX.5 – Mr Ho

NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{7}$	$\overline{10}$	$\overline{9}$	$\overline{14}$	$\overline{10}$	$\overline{50}$	

Question 1 (7 Marks)	Commence a NEW writing booklet	Marks
(a)	Rationalise the denominator $\frac{4 + 3\sqrt{5}}{2 - \sqrt{6}}$, leaving your answer in the simplest form.	2
(b)	Sketch the graph of $y = x + 4 - 3$, showing all important features.	2
(c)	Consider the piecewise defined function $f(x) = \begin{cases} \frac{1}{x} & \text{for } x \leq -1 \\ 2^x + 2 & \text{for } x > -1 \end{cases}$	
i.	Draw a neat sketch of the graph of $y = f(x)$ on one-third of a page in your writing booklet.	2
ii.	Hence state the range of the function, using interval notation.	1

Question 2 (10 Marks)	Commence a NEW writing booklet	Marks
(a)	Differentiate $y = \frac{11}{x^5}$, writing your answer without a negative index.	2
(b)	Find the equation of the normal to the curve $y = 6 + 8x - 3x^4$ at $x = 1$.	3
(c)	Evaluate: $\lim_{x \rightarrow \infty} \frac{3 - x - 2x^2}{5x^2 - 7x + 4}$.	2
(d)	Differentiate $f(x) = 6 - 9x^2$ from first principles.	3

Question 3 (9 Marks)

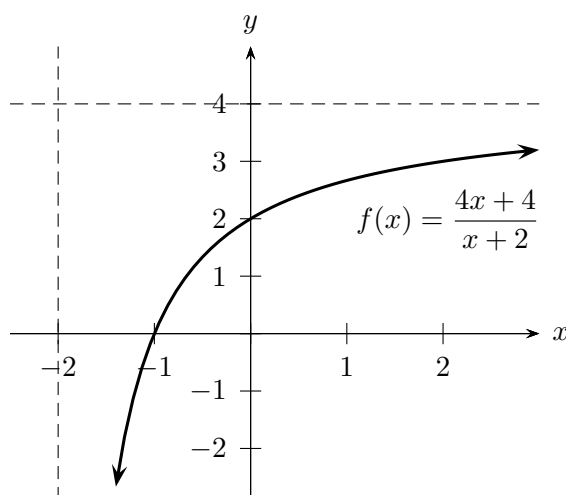
Commence a NEW writing booklet

Marks

- (a) Consider the functions $f(x) = x - 3$ and $g(x) = 2x^2 + 1$.

- i. Find $h(x)$, where $h(x) = g(f(x))$. **1**
- ii. Find the largest domain containing only positive values of x , for which $h(x)$ has an inverse function. **1**
- iii. Hence find the inverse function of $h(x)$. **2**

- (b) The diagram below shows the graph of the curve $y = f(x)$ where $f(x) = \frac{4x+4}{x+2}$ for $x > -2$.



On separate diagrams, sketch the graphs of the following curves, showing clearly the intercepts with the axes and equations of any asymptotes.

It is recommended you draw or trace a copy of $y = f(x)$ on to your writing booklet to use as guidance to demonstrate your understanding.

- i. $y = |f(x)|$. **1**
- ii. $y^2 = f(x)$. **2**
- iii. $y = \frac{1}{f(x)}$. **2**

Question 4 (14 Marks) Commence a NEW writing booklet **Marks**

(a) Fully simplify: $9\cos^2 x + (2 - 3\sin x)^2$. **2**

(b) Find the exact value for $\tan(\alpha + \beta)$ given that $\cos \alpha = \frac{\sqrt{3}}{4}$ for $-\frac{\pi}{2} < \alpha < 0$ and $\sin \beta = \frac{\sqrt{2}}{2}$ for $0 < \beta < \frac{\pi}{2}$. **2**

(You do not need to express your answer with a rational denominator)

(c) Consider the trigonometric function: **3**

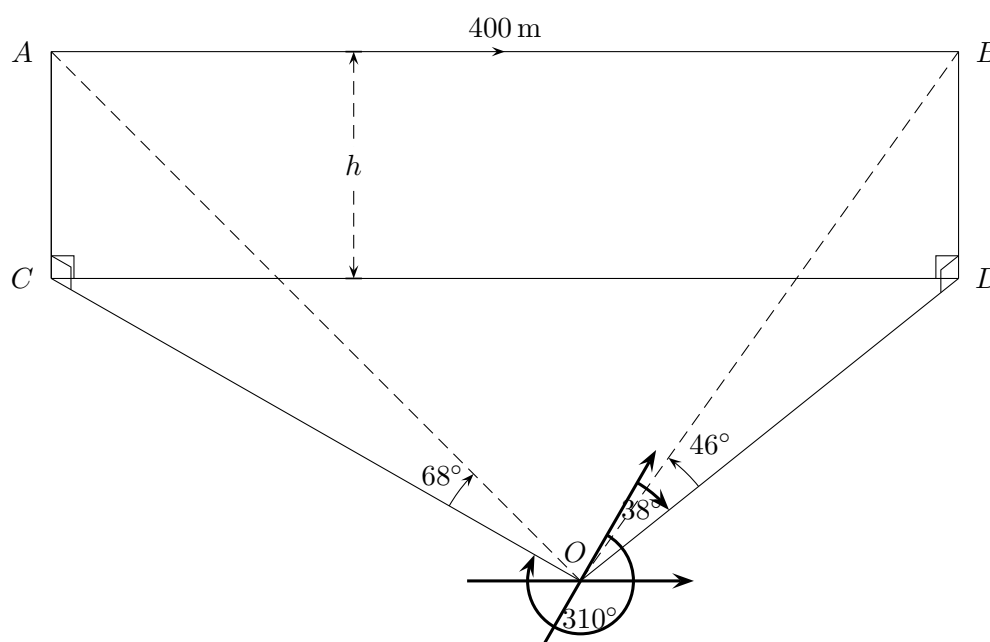
$$y = \sin(3x - \pi) + 1$$

State the necessary transformations, in the correct order, that are need to be applied to $y = \sin x$ to graph the above trigonometric function.

(d) Prove the following trigonometric identity: **2**

$$\frac{\sin 2x + \sin x}{1 + \cos 2x + \cos x} = \tan x$$

(e) From point O on the ground, Mannan observes a drone flying horizontally from point A at a bearing of 310°T and angle of elevation is 68° . After flying 400 m, the drone reaches point B and the bearing changes to 038°T with an angle of elevation of 46° .



i. Show that $OC = h \tan 22^\circ$. **1**

ii. Show that the height h of the drone is given by **3**

$$h = \frac{400}{\sqrt{\tan^2 22^\circ + \tan^2 44^\circ - 2 \tan 22^\circ \tan 44^\circ \cos 88^\circ}}$$

iii. Hence calculate the height of the drone correct to 2 decimal places. **1**

Question 5 (10 Marks)	Commence a NEW writing booklet	Marks
(a) Differentiate with respect to x :	$f(x) = \frac{3x^2 + 2}{x + 1}$	2
(b) Find the coordinates of the stationary point of the function below, and classify its nature with suitable working out.	$y = x^2 - \frac{2}{x}$	3
(c) A water tank is being filled at a variable rate. The height of water h cm at any time t minutes, is given by	$h(x) = 2t^2 - 4t + 50$	
i. Find the initial height of water in the tank.		1
ii. Find the instantaneous rate of change of the height of water when $t = 4$ minutes.		2
iii. Find the average rate at which the height has changed over the first 4 minutes.		2

End of paper.

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NSW Education Standards Authority

2020 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

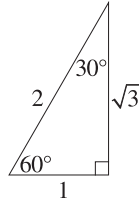
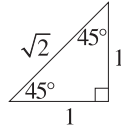
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

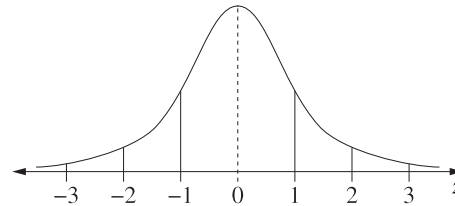
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x_1\underline{i} + y_1\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Solutions to Year 11 Extension 1 Task 3

Question 1:

(a)

$$\frac{4 + 3\sqrt{5}}{2 - \sqrt{6}} \times \frac{2 + \sqrt{6}}{2 + \sqrt{6}}$$

$$= \frac{(4 + 3\sqrt{5})(2 + \sqrt{6})}{4 - 6}$$

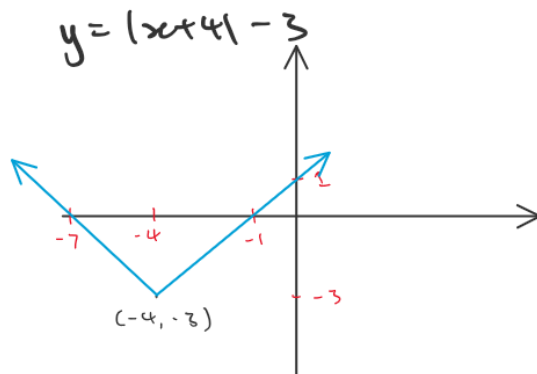
① Mark for rationalising correctly

$$= \frac{8 + 4\sqrt{6} + 6\sqrt{5} + 3\sqrt{30}}{-2}$$

$$= - \frac{8 + 4\sqrt{6} + 6\sqrt{5} + 3\sqrt{30}}{2}$$

① Mark for simplest form

(b)



① Mark for correct vertex

① Mark for correct graph.

→ Find the x-intercept by letting $y=0$

$$0 = |x+4| - 3$$

$$|x+4| = 3$$

$$x+4 = 3$$

$$x = -1$$

$$x+4 = -3$$

$$x = -7$$

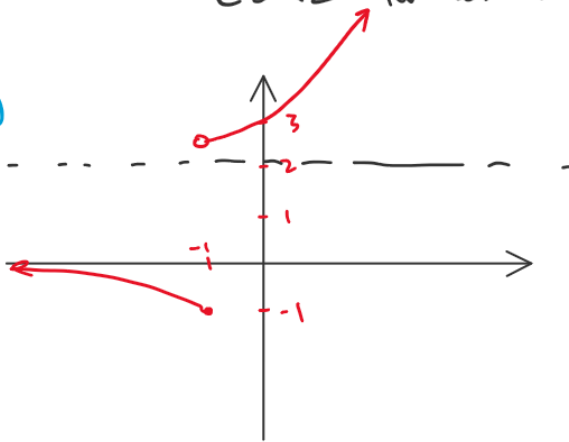
→ Find the y-intercept by letting $x=0$

$$y = |4| - 3$$

(c) $y = 1$

$$f(x) = \begin{cases} \frac{1}{x} & \text{for } x \leq -1 \\ 2^x + 2 & \text{for } x > -1 \end{cases}$$

(i)



① Mark for indicating correct domain restrictions

① Mark for correct graph and labelling of intercepts

(ii) $R: y \in [-1, 0) \cup [2.5, \infty)$

Question 2.

(a)

$$y = 11x^{-5}$$

$$\frac{dy}{dx} = -55x^{-6} \quad \checkmark$$

① Mark for correct derivative

$$= -\frac{55}{x^6} \quad \checkmark$$

① Mark for answer without negative index.

(b)

$$y = 6 + 8x - 3x^4$$

$$\frac{dy}{dx} = 8 - 12x^3$$

$$\text{at } x=1:$$

$$m = 8 - 12(1)^3$$

$$m_1 = -4$$

$$m_2 = \frac{1}{4}$$

① Mark for correct derivative.

① Mark for correct gradient

substitute $x=1$ into

y for y -coordinate: $y=11$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 11 = \frac{1}{4}(x - 1)$$

$$y - 11 = \frac{1}{4}x - \frac{1}{4}$$

$$y = \frac{1}{4}x + \frac{43}{4}$$

① Mark for correct equation

(c)

$$\lim_{x \rightarrow \infty} \frac{3 - x - 2x^2}{5x^2 - 7x + 4} \times \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3}{x^2} - \frac{1}{x} - 2}{5 - \frac{7}{x} + \frac{4}{x^2}}$$

① correct simplification

$$= \frac{0 - 0 - 2}{5 - 0 + 0}$$

$$= -\frac{2}{5}$$

① correct answer.

(d)

$$f(x) = 6 - 9x^2$$

$$f(x+h) = 6 - 9(x+h)^2$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{6 - 9(x+h)^2 - (6 - 9x^2)}{h}$$

✓ (i) Mark for correct substitution

$$\lim_{h \rightarrow 0} \frac{6 - 9(x^2 + 2xh + h^2) - 6 + 9x^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{6} - \cancel{9x^2} - (18xh - 9h^2 - \cancel{6} + \cancel{9x^2})}{h}$$

✓ (i) Mark for correct simplifying

$$\lim_{h \rightarrow 0} \frac{-18xh - 9h^2}{h}$$

$$\lim_{h \rightarrow 0} \frac{-9\cancel{h}(2x + h)}{\cancel{h}}$$

$$= -18x$$

✓ (i) Mark for correct answer.

Question 3

(a)

(i) $f(x) = x - 3$

$$g(x) = 2x^2 + 1$$

$$g(f(x)) = 2(x-3)^2 + 1$$

$$g(f(x)) = 2(x^2 - 6x + 9) + 1$$

$$g(f(x)) = 2x^2 - 12x + 18 + 1$$

$$g(f(x)) = 2x^2 - 12x + 19$$

$$\therefore h(x) = 2x^2 - 12x + 19$$

① Mark for correct function.

(ii)

$$x = -\frac{b}{2a}$$

$$x = \frac{12}{4}$$

$$x = 3$$

→ "largest positive domain"

$$\therefore x \geq 3$$

① Mark for correct domain

(iii)

$$h(x) = 2(x-3)^2 + 1$$

$$x = 2(y-3)^2 + 1$$

① Mark for correct change in "x" and "y"

$$2(y-3)^2 = x - 1$$

$$(y-3)^2 = \frac{x-1}{2}$$

$$y-3 = \sqrt{\frac{x-1}{2}} \quad (y \geq 3)$$

$$y = 3 + \sqrt{\frac{x-1}{2}}$$

$$\therefore h^{-1}(x) = \sqrt{\frac{x-1}{2}} + 3$$

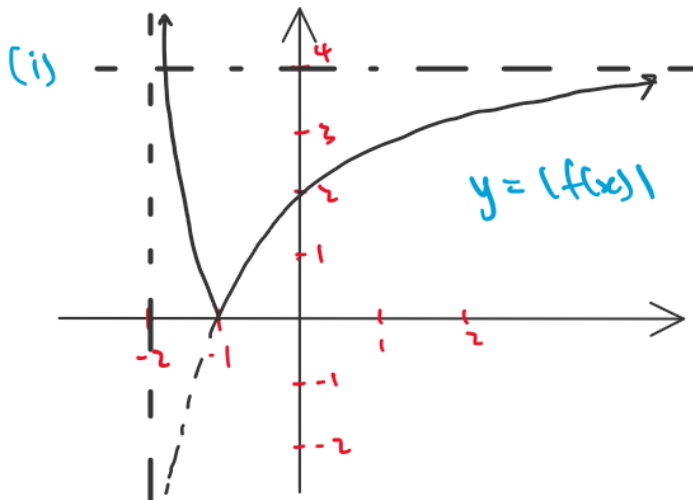
✓ ① Mark for correct inverse function

(b)

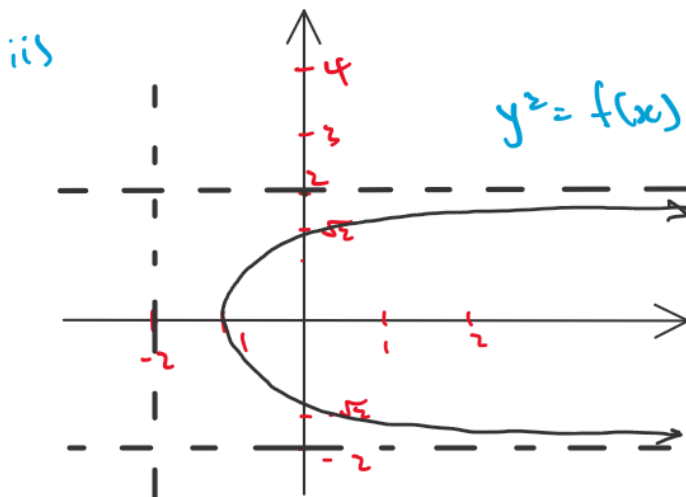
$$y = f(x)$$

$$f(x) = \frac{4x+4}{x+2}$$

for $x > -2$



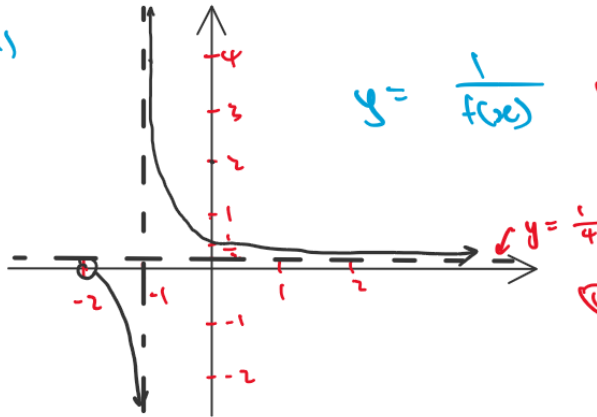
① Mark for correct graph



① Mark for correct intercepts

① Mark for correct graph

(iii)



$$y = \frac{1}{f(x)}$$

① Mark for correct intercepts

① Mark for correct graph.

Question 4.

(a)

$$\begin{aligned} & 9 \cos^2 x + (2 - 3 \sin x)^2 \\ &= 9 \cos^2 x + (4 - 12 \sin x + 9 \sin^2 x) \\ &= 9 \cos^2 x + 9 \sin^2 x + 4 - 12 \sin x \\ &= 9 (\cos^2 x + \sin^2 x) + 4 - 12 \sin x \\ &= 13 - 12 \sin x \end{aligned}$$

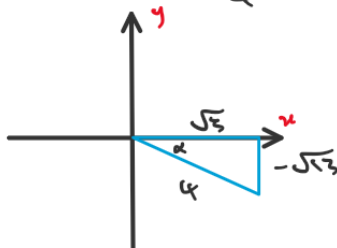
① Correct expansion

① Correct answer.

(b)

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

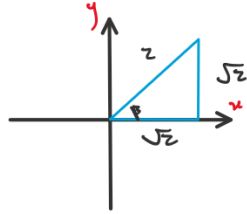
$$\text{Given that } \cos \alpha = \frac{\sqrt{3}}{4} \text{ for } -\frac{\pi}{2} < \alpha < 0$$



$$\tan \alpha = -\frac{\sqrt{3}}{\sqrt{3}}$$

① Mark for the correct $\tan \alpha$ and $\tan \beta$ solutions

$$\text{Given that } \sin \beta = \frac{\sqrt{2}}{2} \text{ for } 0 < \beta < \frac{\pi}{4}$$



$$\tan \alpha = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

① Mark for the correct answer.

$$\begin{aligned} \therefore \tan(\alpha + \beta) &= \frac{-\frac{\sqrt{3}}{\sqrt{3}} + 1}{1 + \frac{\sqrt{3}}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{-\sqrt{3} + \sqrt{3}}{\sqrt{3} + \sqrt{3}} \end{aligned}$$

cc)

$$y = \sin(3x - \alpha) + 1$$

1. Factorise at 3:

$$y = \sin\left[3\left(x - \frac{\alpha}{3}\right)\right] + 1$$

$$\text{period} = \frac{2\pi}{n}$$

$$= \frac{2\pi}{3} = \text{horizontal stretch by a factor of } \frac{1}{3}$$

① Mark for correct transformation terminology.

2. Consider the phase shift:

the function moves $\frac{\alpha}{3}$ units to the right.

① Mark for correct phase shift

3. Consider the vertical upwards shift of 1.

∴ 1. Horizontal stretch by a factor of $\frac{1}{3}$.

2. Phase shift = $\frac{\pi}{3}$

3. Vertical shift = 1 up. (1) Mark for correct vertical shift.

(d)

$$\frac{\sin 2x + \sin x}{1 + \cos 2x + \cos x} = \tan x$$

$$\begin{aligned}\star \sin 2x &= 2 \sin x \cos x \\ \cos 2x &= 2 \cos^2 x - 1\end{aligned}$$

L.H.S:

$$= \frac{2 \sin x \cos x + \sin x}{1 + 2 \cos^2 x - 1 + \cos x}$$

✓ (1) Mark for changing $\sin 2x$ and $\cos 2x$

$$= \frac{\sin x (\cancel{2 \cos x + 1})}{\cos x (\cancel{2 \cos x + 1})}$$

$$= \frac{\sin x}{\cos x}$$

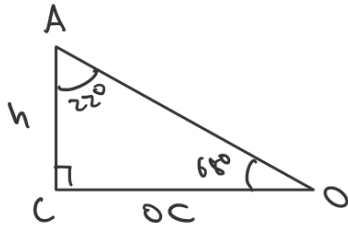
✓ (1) Mark for correct simplification

$$= \tan x$$

$$= \text{R.H.S}$$

(e)

(i)



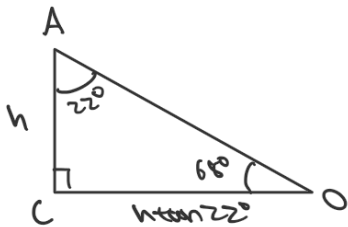
$$\tan 22^\circ = \frac{OC}{h}$$

$$OC = h \tan 22^\circ$$

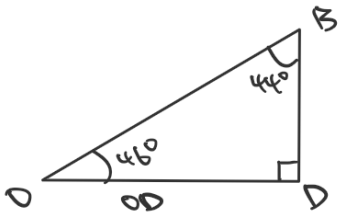
① Mark for correct trigonometric ratio.

(ii)

Draw three 2-D triangles:



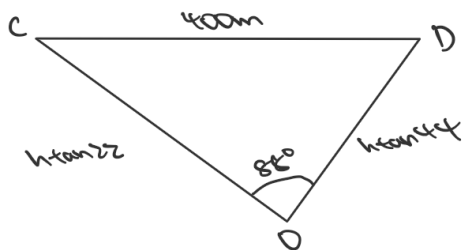
$$OC = h \tan 22^\circ$$



$$\tan 44^\circ = \frac{OD}{h}$$

$$OD = h \tan 44^\circ$$

① Mark for correct side of OD.



$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$400^2 = (h \tan 22^\circ)^2 + (h \tan 44^\circ)^2 - (2 \times h \tan 22^\circ \times h \tan 44^\circ \times \cos 86^\circ) \quad \checkmark$$

$$400^2 = h^2 \tan^2 22^\circ + h^2 \tan^2 44^\circ - 2h \tan 22^\circ \tan 44^\circ \cos 86^\circ$$

① Mark for correct substitution

$$400^2 = h^2 (\tan^2 22^\circ + \tan^2 44^\circ - 2 \tan 22^\circ \tan 44^\circ \cos 86^\circ)$$

$$h^2 = \frac{400^2}{\tan^2 22^\circ + \tan^2 44^\circ - 2 \tan 22^\circ \tan 44^\circ \cos 86^\circ} \quad \checkmark$$

① Mark for correct simplification.

$$h = \frac{400}{\sqrt{\tan^2 22^\circ + \tan^2 44^\circ - 2 \tan 22^\circ \tan 44^\circ \cos 86^\circ}}$$

(iii)

$$h = 386.96 \text{ m.}$$

(2 d.p.) $\checkmark$

① Mark for correct answer.

Question 5.

(a)

$$f(x) = \frac{3x^2 + 2}{x + 1}$$

$$u = 3x^2 + 2 \quad v = x + 1$$

$$u' = 6x \quad v' = 1$$

$$f'(x) = \frac{6x^2 + 6x - (3x^2 + 2)}{(x+1)^2} \quad \checkmark \quad \textcircled{1} \text{ Mark for correct use of quotient rule}$$

$$= \frac{6x^2 + 6x - 3x^2 - 2}{(x+1)^2}$$

$$= \frac{3x^2 + 6x - 2}{(x+1)^2} \quad \checkmark \quad \textcircled{1} \text{ Mark for correct simplification}$$

(6)

$$y = x^2 - 2x^{-1}$$

$$\frac{dy}{dx} = 2x + 2x^{-2}$$

$$= 2x + \frac{2}{x^2}$$

$\checkmark$ $\textcircled{1}$ correct first derivative

Let $\frac{dy}{dx} = 0$ to find stationary point.

$$-2x = \frac{2}{x^2}$$

$$-2x^3 = 2$$

$$x^3 = -1$$

$$x = -1 \leftarrow \text{substitute into } y \text{ for y-coordinate.}$$

$$y = 3$$

Sub stationary point into second derivative to determine nature.

$$\frac{d^2y}{dx^2} = 2 - 4x^{-3}$$

$$= 2 - \frac{4}{x^3}$$

sub in $x = -1$

$$\frac{d^2y}{dx^2} = 2 - \frac{4}{(-1)^3}$$

$$= 2 + 4$$

$$= 6$$

$\therefore$ minimum

$\therefore (-1, 3)$ is a minimum

✓ ① Mark for correct
concavity
✓ ① Mark for correct
coordinates

(c)

For $h(x) = 2x^2 - 4x + 50$

(i) let $x = 0$

$$h(0) = 50 \text{ cm}$$

✓ ① Mark for correct answer

(ii) For rate of change, find
the gradient function:

$$h'(x) = 4x - 4$$

$$h'(4) = 16 - 4$$

$$= 12 \text{ cm/min}$$

✓ ① Mark for correct gradient
function

✓ ① Mark for correct rate with
units

(iii) $h(0) = 50$

$$h(4) = 66$$

✓ ① Mark for $h(4)$

$$\therefore \frac{66 - 50}{4 - 0}$$

$$= \frac{16}{4}$$

$\therefore$ 4 cm/min for ✓ ① Mark for correct
average rate of change. ~~answer~~ with substitution.